

IT4010 - Research Project

Project ID: 25-26J-157

BUSINESS MODEL CANVAS

DECISION SUPPORT SYSTEM FOR AGARWOOD CULTIVATION IN SRI LANKA

BUSINESS MODEL CANVAS – DECISION SUPPORT SYSTEM FOR AGARWOOD CULTIVATION



KEY PARTNERS

- Pintanna Plantations (pilot / industry collaboration)
- Sri Lankan agricultural research bodies (Department of Forestry, universities)
- Technology providers (IoT hardware vendors, blockchain platforms)
- Export associations and government regulators
- International fragrance and pharmaceutical companies (buyers)



KEY ACTIVITIES

- Development and maintenance of IoT + dashboard platform
- Sensor deployment and tree tagging
- Data collection, analytics, and visualization
- Blockchain integration for certification and traceability
- AI model training for resin detection and growth classification
- Customer support, training programs, and pilot deployments



KEY RESOURCES

- IoT sensors, QR/NFC tags, GPS systems
- Cloud infrastructure and blockchain platform
- AI/ML models for image recognition and resin prediction
- Development & support team (software engineers, IoT experts, agronomists)
- Partnerships with agricultural institutions and certification authorities



VALUE PROPOSITIONS

- Data-driven decision support system that increases agarwood resin yield and quality
- Reduction of losses caused by premature or misinformed harvesting
- Transparent blockchain-backed certification and traceability to attract international buyers
- IoT-enabled monitoring and prediction to optimize irrigation, inoculation, and operations
- Real-time virtual tree map for efficient worker management and route optimization
- AI-based resin detection for higher accuracy in harvest readiness decisions



CUSTOMER RELATIONSHIPS

- Long-term contracts with plantations for system deployment and maintenance
- Dedicated technical support and on-site training for farmers and workers
- Periodic software/hardware updates and upgrades
- Community forums and advisory support for best practices in agarwood cultivation



CHANNELS

- Direct sales to plantation owners and cooperatives
- Partnerships with agro-tech distributors and government agriculture agencies
- Online platform/dashboard (subscription or license-based access)
- Mobile app for workers (QR/NFC tree tracking, alerts, updates)
- Web-based buyer dashboard for certification and traceability



CUSTOMER SEGMENTS

- Large-scale agarwood plantation owners
- Smallholder farmers interested in agarwood cultivation
- Agricultural investors and exporters
- Certification bodies, regulators, and government forestry agencies
- International buyers of agarwood resin (perfume, incense, pharmaceuticals)



COST STRUCTURE

- Hardware: IoT sensors, NFC/QR tags, GPS devices
- Cloud hosting, data storage, and blockchain transaction costs
- Software development and ongoing maintenance
- Field deployment, training, and pilot costs
- Marketing, commercialization, and certification partnership expenses
- Customer support and system upgrades



REVENUE STREAMS

- One-time setup fee for IoT devices, tags, and system installation
- Subscription-based SaaS for dashboard, analytics, and blockchain certification
- Premium features (AI image analysis, resin scoring models, route optimization)
- Consulting services (plantation digitization, training, certification guidance)
- Commission or licensing fees from buyer-facing certification dashboards